CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM) **Sem - I University Examination, 2009-10**

ME – 702 Advances in Manufacturing Technology

Date: 23.12.2009, Wednesday **Time: 01:00 p.m. to04:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.
3. Use of PSG Design data book is permitted.

SECTION- I

Q1	**(a)**	What are the parameters to be controlled in the resistance welding process?	**03**
	(b)	Explain the process of metal transfer in GMAW process with neat sketches.	**04**
	(c)	Explain the principle of keyhole welding. Discuss the effect of horizontal forces on the keyhole wall.	**04**
Q2	**(a)**	What is EPC in reference to RFID system?	**02**
	(b)	State the merits and demerits of various RFID frequencies.	**04**
	(c)	Discuss the process of fibre drawing using Chemical Vapour Deposition Process with a neat sketch.	**06**

OR

Q2	**(a)**	Why is sputter deposition so much slower than evaporation deposition? Make a detailed comparison of the two deposition methods.	**06**
	(b)	Discuss the classes, types and designs of RFID tags	**06**
Q3	**(a)**	Enlist the factors required to choose a simulation software	**02**
	(b)	Enumerate the benefits of simulation modeling and analysis	**04**
	(c)	Elaborate the steps in developing a simulation model	**06**

OR

Q3		Write Short Notes on the following:-	
	(a)	Micro Machining	**04**
	(b)	Deep Hole Drilling	**04**
	(c)	Hot Machining	**04**

SECTION - II

Q4 (a) Differentiate between spontaneous and stimulated emission in Laser beam machining. **03**

(b) Find out the approximate time required to machine an aperture of diameter equal to 3.0 mm in a tungsten carbide plate (fracture hardness = 6900 N/mm^2) of thickness equal to one and half times of hole diameter. The mean abrasive grain size is 0.010 mm diameter. The feed force is equal to 2.8 N. The amplitude of tool oscillation is 25 micron and the frequency is equal to 25 kHz. The tool material used is copper having fracture hardness equal to 1.5×10^3 N/mm^2. The slurry contains one part abrasive to one part of water. Take the values of different constants as $K_1 = 0.3$, $K_2 = 1.8\ mm^2$, $K_3 = 0.6$ and abrasive density = 3.8 gm/cm^3. **08**

Also calculate the ratio of the volume removed by throwing mechanism to the volume removed by hammering mechanism.

Q5 (a) What is regression model? Explain the type of regression analysis used for two or more than two variables. **06**

(b) Following is the multi city annual sales data for car manufacturers' i. e. Hyundai, TATA and General Motors, each sold in 4 cities. **06**

Cities	Product Sales (in Thousands)		
	Hyundai	TATA	General Motors
Delhi	68	45	29
Bangalore	36	51	47
Mumbai	46	61	34

Setup a table of analysis of variance and calculate F. State whether the difference between the yields of two varieties is significant, taking 7.71 as the value of F at 5% level for $V_1 = 1$ and $V_2 = 4$.

OR

Q5 (a) Discuss the expected benefits of Virtual manufacturing. **02**

(b) Discuss the role of Process models in Virtual Manufacturing? **04**

(c) Give an overview on Control hierarchy of a virtual manufacturing cell. **06**

Q6 (a) Write short notes on: **06**

1. Explosives used in Explosive Forming
2. Advantages, limitations and application of Hydro forming

(b) Explain the framework of Virtual manufacturing system showing the structure of virtual manufacturing environment. **06**

OR

Q6 (a) Explain the grain throwing model proposed by Shaw for mechanics of cutting during ultrasonic machining. **06**

(b) What are different spark generators used in EDM? Explain working of R-C circuit in EDM process. Describe the effect of Resistance and Capacitance on MRR. **06**